Scientific Notation

Scientific Notation (also called Standard Form in Britain) is a special way of writing numbers:

Like this: 700 → 7×10^2

A Number — In Scientific Notation

Or this: 4,900,000,000 → 4.9×10^9

A Number — In Scientific Notation

It makes it easy to use big and small values.

OK, How Does it Work?

Example: 700

Why is 700 written as $\mathbf{7 \times 10^2}$ in Scientific Notation ?

$700 = 7 \times 100$

and $100 = 10^2$ *(see powers of 10)*

so $700 = \mathbf{7 \times 10^2}$

Both **700** and $\mathbf{7 \times 10^2}$ have the same value, just shown in different ways.

Example: 4,900,000,000

$1{,}000{,}000{,}000 = 10^9$,
so $4{,}900{,}000{,}000 = \mathbf{4.9 \times 10^9}$ in Scientific Notation

The number is written in **two parts**:

Just the **digits**, with the decimal point placed after the first digit, followed by

× 10 to a power that puts the decimal point where it should be
(i.e. it shows how many places to move the decimal point).

Digits Power of 10

$$5326.6 = 5.3266 \times 10^{3}$$

A Number In Scientific Notation

In this example, 5326.6 is written as **5.3266×10^3**,
because $5326.6 = 5.3266 \times 1000 = 5.3266 \times 10^3$

Try It Yourself

Enter a number and see it in Scientific Notation:

Enter Number: Example

3.1416

Scientific Notation:

3.1416×10^0

© 2015 MathsIsFun.com v 0.85

Now try to use Scientific Notation yourself:

Scientific Notation:

Example 3.1416 **× 10** 2

Number: **314.16**

© 2015 MathsIsFun.com v 0.85

Other Ways of Writing It

We can use the **^** symbol (above the 6 on a keyboard), as it is easy to type.

3.1 × 10^8

Example: **3 × 10^4 is the same as 3 × 10^4**

3 × 10^4 = 3 × 10 × 10 × 10 × 10 = 30,000

Calculators often use "E" or "e" like this:

Example: **6E+5 is the same as 6 × 10^5**

6E+5 = 6 × 10 × 10 × 10 × 10 × 10 = 600,000

Example: **3.12E4 is the same as 3.12 × 10^4**

3.12E4 = 3.12 × 10 × 10 × 10 × 10 = 31,200

How to Do it

To figure out the power of 10, think "how many places do I move the decimal point?"

When the number is 10 or greater, the decimal point has to move **to the left**, and the power of 10 is **positive**.

When the number is smaller than 1, the decimal point has to move **to the right**, so the power of 10 is **negative**.

Example: 0.0055 is written $\mathbf{5.5 \times 10^{-3}}$

Because $0.0055 = 5.5 \times 0.001 = 5.5 \times 10^{-3}$

Example: 3.2 is written $\mathbf{3.2 \times 10^{0}}$

We didn't have to move the decimal point at all, so the power is $\mathbf{10^{0}}$

But it is now in Scientific Notation

Check!

After putting the number in Scientific Notation, just check that:

The "digits" part is between 1 and 10 (it can be 1, but never 10)

The "power" part shows exactly how many places to move the decimal point

Why Use It?

Because it makes it easier when dealing with very big or very small numbers, which are common in Scientific and Engineering work.

Example: it is easier to write (and read) 1.3×10^{-9} than 0.0000000013

It can also make calculations easier, as in this example:

Example: a tiny space inside a computer chip has been measured to be 0.00000256m wide, 0.00000014m long and 0.000275m high.

What is its volume?

Let's first convert the three lengths into scientific notation:

width: 0.000 002 56m = 2.56×10^{-6}

length: 0.000 000 14m = 1.4×10^{-7}

height: 0.000 275m = 2.75×10^{-4}

Then multiply the digits together (ignoring the ×10s):

2.56 × 1.4 × 2.75 = 9.856

Last, multiply the ×10s:

$10^{-6} \times 10^{-7} \times 10^{-4} = 10^{-17}$ (easier than it looks, just **add −6, −4 and −7** together)

The result is **9.856×10^{-17} m^3**

It is used a lot in Science:

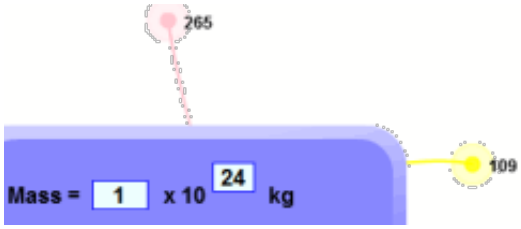
Example: Suns, Moons and Planets

The Sun has a Mass of 1.988×10^{30} kg.

Easier than writing 1,988,000,000,000,000,000,000,000,000,000 kg (and that number gives a false sense of many digits of accuracy.)

Play With It!

Use Scientific Notation in Gravity Freeplay



It can also save space! Here is what happens when you double on each square of a chess board:

1	2	4	8	16	32	64	128
256	512	1×10^{3}	2×10^{3}	4×10^{3}	8×10^{3}	2×10^{4}	3×10^{4}
7×10^{4}	1×10^{5}	3×10^{5}	5×10^{5}	1×10^{6}	2×10^{6}	4×10^{6}	8×10^{6}
2×10^{7}	3×10^{7}	7×10^{7}	1×10^{8}	3×10^{8}	5×10^{8}	1×10^{9}	2×10^{9}
4×10^{9}	9×10^{9}	2×10^{10}	3×10^{10}	7×10^{10}	1×10^{11}	3×10^{11}	5×10^{11}
1×10^{12}	2×10^{12}	4×10^{12}	9×10^{12}	2×10^{13}	4×10^{13}	7×10^{13}	1×10^{14}
3×10^{14}	6×10^{14}	1×10^{15}	2×10^{15}	5×10^{15}	9×10^{15}	2×10^{16}	4×10^{16}
7×10^{16}	1×10^{17}	3×10^{17}	6×10^{17}	1×10^{18}	2×10^{18}	5×10^{18}	9×10^{18}

Values are rounded off, so 53,6870,912 is shown as just 5×10^{8}

That last value, shown as 9×10^{18} is actually 9,223,372,036,854,775,808

Engineering Notation

Engineering Notation is like Scientific Notation, except that we only use powers of ten that are multiples of 3 (such as 10^{3}, 10^{-3}, 10^{12} etc).

Examples:

2,700 is written **2.7×10^3**

27,000 is written **27×10^3**

270,000 is written **270×10^3**

2,700,000 is written **2.7×10^6**

Example: 0.00012 is written **120×10^{-6}**

Notice that the "digits" part can now be between 1 and 1,000 (it can be 1, but never 1,000).

The advantage is that we can replace the **×10**s with Metric Numbers. So we can use standard words (such as thousand or million), prefixes (such as kilo, mega) or the symbol (k, M, etc)

Example: 19,300 metres is written **19.3×10^3 m, or 19.3 km**

Example: 0.00012 seconds is written **120×10^{-6} s, or 120 microseconds**

Question 1 Question 2 Question 3 Question 4 Question 5
Question 6 Question 7 Question 8 Question 9 Question 10
Question 11 Question 12

Copyright © 2020 MathsIsFun.com